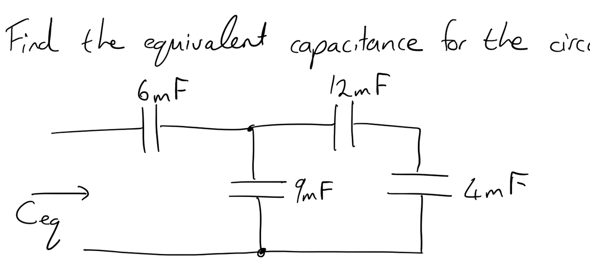
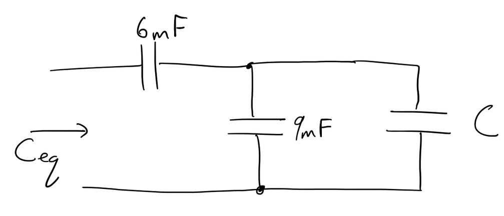
Problem class 2

Question 1

Find the equivalent capacitance for the circuit:

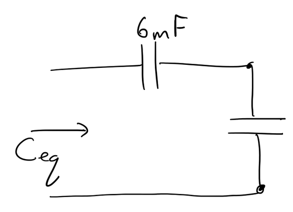


⇕



$C_1 = \dfrac{1}{\frac{1}{9mF} + \frac{1}{12mF}} = 3mF$

⇕

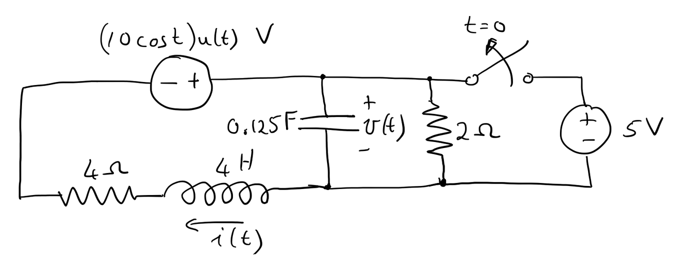


$C_2 = 3mF + 9mF = 12mF$

$C_{eq} = \dfrac{1}{\frac{1}{6mF} + \frac{1}{12mF}} = 4mF$

Question 2

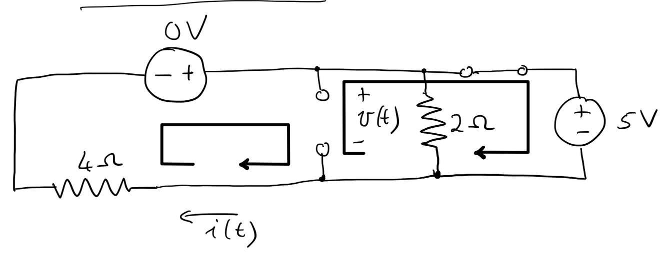
A power supply system is modelled by the circuit:



Compute an expression for $v(t)$, $t \geq 0$

(a) Compute initial conditions

Circuit for $t < 0$

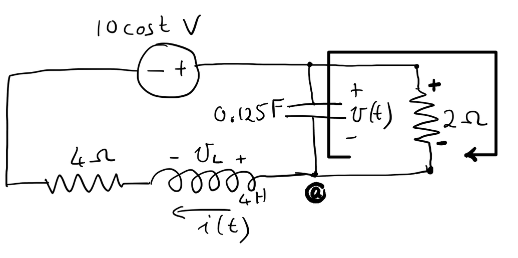


$v(0^-)$ = KVL : $5V - v(0^-) = 0$ ∴ $v(0^-) = 5V$

$i(0^-)$ = KVL : $0V - v(0^-) - i(0^-) \cdot 4\Omega = 0$ ∴ $i(0^-) = -\frac{5}{4}A$

(b) Derive the differential equation for v:

Circuit for $t > 0$



KCL at ⓐ: $i = i_c + i_R$ (1)

$i_R = \frac{v_R}{2\Omega} = \frac{1}{2}v(t)$ (2)

$i_c = C\frac{dv}{dt} = 0.125\frac{dv}{dt}$ (3)

⇒ $i = \frac{1}{2}v + \frac{1}{8}\frac{dv}{dt}$ (4)

KVL around left loop: $10\cos t - v - v_L - 4i = 0$ (5)

$v_L = L\frac{di}{dt}$ (6)

Substitute (4) and (6) into (5) to get

$\frac{dv}{dt} + A\frac{di}{dt} + Bv = v_f$ (7)

$A = 8$ $B = 6$ v_f: $20\cos(t)$

Substitute (4) into (7) to get

$\frac{d^2v}{dt^2} + C\frac{dv}{dt} + Bv = v_f$ (8)

$C = 5$

∴ $\frac{dv}{dt} + 5\frac{dv}{dt} + 6v = 20\cos(t)$

(c) The characteristic equation is: $\lambda^2 + 5\lambda + 6 = 0$

The roots of the characteristic equation are: $\lambda = -2$ and -3

(d) Recall from part (a) that

$v(0^-) = 5V$

$i(0^-) = -\frac{5}{4}A$

and from (4):

$i(t) = \frac{1}{2}v + \frac{1}{8}\frac{dv}{dt}$

⇒ Initial conditions of (8) are:

$v(0^+) = 5V$ (9)

$\frac{dv}{dt}(0^+) = -30V/s$ (10)

(e) The solution to (8) has the form

1) $v(t) = De^{-3t} + Ee^{-2t}$

2) $v(t) = De^{-3t} + Ee^{-2t} + F\sin t + G\cos t$

3) $v(t) = F\sin t + G\cos t$

4) None of the above

The particular solution has to satisfy the ODE (8)

⇒ $v_p(t) = Ae^{-2t} + Be^{-3t} = 45e^{-2t} - 40e^{-3t}$

The general solution also has to satisfy the ODE (8) $V_c(t) = C\cos(t) + D\sin(t) = 2\cos(t)$

The initial conditions (9) and (10) could be used to compute values for D and E so that

⇒ $v(t) = 45e^{-2t} - 40e^{-3t} + 2\cos(t)$